An Attention-Enhanced EfficientNet-B3 Framework for Robust DeepFake Detection on Compressed Social Media Images

Arnav Dutta[†]
Computer Science and Engineering
GITA Autonomous College
Bhubaneswar, India
arnavdutta5282@gmail.com

Ansuman Sutar[†]
Computer Science and Engineering
GITA Autonomous College
Bhubaneswar, India
ansumansutar@gmail.com

Anshuman Patnaik[†]
Computer Science and Engineering
GITA Autonomous College
Bhubaneswar, India
anshumanpatnaik@gmail.com

Arjya Pratap Nayak[†]
Computer Science and Engineering
GITA Autonomous College
Bhubaneswar, India
arjyapratapnayak@gmail.com

***Abstract*—Recent advances in generative networks allow the creation of photorealistic synthetic faces that threaten digital trust and information integrity. Standard convolutional architectures such as EfficientNet-B3 deliver strong accuracy on clean benchmarks, yet suffer significant performance drops under lossy social media compression. Transmission pipelines introduce Discrete Cosine Transform quantization noise that erases subtle forensic artifacts while generating deceptive block boundary edges. To address this issue, we introduce Coord-EfficientNet-B3, an attention enhanced framework combining Coordinate Attention with stochastic compression aware training. By factorizing spatial pooling into orthogonal horizontal and vertical coordinate representations, the model isolates structural tampering cues while ignoring two dimensional compression artifacts. Evaluated across 12,000 multi-quality factor inferences, the proposed architecture reduces overall compressed errors by 14.4% and suppresses false alarms by 24.2% relative to the unaugmented baseline ($p = 0.00116$, McNemar test). On an unseen holdout of face portraits, the model lowers total errors by 49.1% and halves false negatives at the standard threshold.**

***Index Terms*—DeepFake detection, EfficientNet-B3, Coordinate Attention, lossy compression, image forensics, false negatives.**

I. Introduction

The rapid rise of generative artificial intelligence has made it straightforward to produce realistic synthetic and manipulated imagery. Modern Generative Adversarial Networks such as StyleGAN, along with contemporary neural diffusion models, can synthesize human facial portraits featuring natural skin textures, convincing lighting, and authentic expressions [1], [2]. Although these synthesis technologies offer promising opportunities in digital entertainment and creative arts, they present serious risks when weaponized for impersonation, identity fraud, and political disinformation [3]. Establishing robust forensic methods capable of verifying digital authenticity in real world dissemination pipelines is therefore vital.

Deep learning architectures based on Convolutional Neural Networks have emerged as the standard foundation for deepfake detection because they extract hierarchical visual representations directly from pixel inputs [4]. EfficientNet architectures offer a balanced combination of computational efficiency and representational capacity through compound scaling. Deepa et al. [5] examined an EfficientNet B3 baseline for facial manipulation detection, demonstrating that it surpasses classical machine learning algorithms such as Random Forests and Support Vector Machines by reaching over 97.9% accuracy on clean benchmark imagery.

Nevertheless, an unresolved operational vulnerability arises when forensic models encounter everyday social media transmissions. Messaging services and social networks routinely apply lossy JPEG compression to reduce storage requirements and bandwidth usage [6]. This process applies Discrete Cosine Transform quantization tables that discard subtle high frequency forensic artifacts where generative traces usually reside [7]. Simultaneously, block boundaries from lossy compression introduce artificial grid ringing that standard convolutional kernels often mistake for genuine manipulation boundaries [8]. Consequently, conventional classifiers trained strictly on pristine images suffer marked increases in false alarms and missed manipulations when tested on compressed content.

To resolve this limitation, we introduce Coord EfficientNet B3, integrating Coordinate Attention [15] into the EfficientNet B3 backbone alongside stochastic compression aware training. Unlike conventional channel attention modules that condense all spatial details into single scalar values [13], Coordinate Attention factorizes spatial pooling into separate horizontal and vertical one dimensional feature vectors. This structural separation allows the network to capture location sensitive forensic cues along one axis while preserving a broad spatial context along the other, preventing feature dispersion across compression block grids.

The primary contributions of this work are summarized as follows:

1) We design Coord-EfficientNet-B3, embedding Coordinate Attention into the final stage of EfficientNet-B3 to capture directional manipulation boundaries with negligible parameter overhead.
2) We introduce a stochastic compression-aware training pipeline that exposes the backbone to randomized JPEG quality factors, learning compression-invariant representations.

[†]These authors contributed equally to this work.

3) Across 12,000 multi-quality factor evaluations, the proposed architecture cuts compressed errors by 14.4% and suppresses false positives by 24.2% ($p = 0.00116$, McNemar test) relative to the baseline EfficientNet-B3 [5].
4) On an unseen holdout of 200 facial portraits, the model reduces total misclassifications by 49.1% and halves false negatives at the standard threshold $\tau = 0.50$.

II. Related Work

A. DeepFake Detection and CNN Backbones

Early forensic methodologies relied primarily on hand-crafted physical and biological anomalies, such as irregular blink frequencies [11], corneal reflection discrepancies, and head pose irregularities [3]. However, as generative architectures matured [2], these explicit physiological imperfections became increasingly imperceptible, compelling researchers to adopt deep representation learning.

Afchar et al. [12] developed MesoNet, utilizing compact convolutional layers to identify mesoscopic facial tampering boundaries. Wang et al. [9] demonstrated that CNN-generated images share common frequency domain artifacts that convolutional models can reliably recognize, though these signatures remain sensitive to post-processing. Chai et al. [10] examined localized patch artifacts, revealing that synthetic imagery retains structural flaws across specific facial components. Tan and Le [4] proposed EfficientNet, which scales depth, width, and resolution uniformly. Building on this backbone, Deepa et al. [5] applied EfficientNet-B3 to deepfake detection, attaining 97.95% accuracy on uncompressed benchmark datasets and confirming that deep representations substantially outperform traditional hand-crafted texture classifiers.

B. Robustness Under Social Media Compression

The vulnerability of deep neural classifiers to image compression is widely acknowledged in multimedia forensics. JPEG compression divides an image into 8×8 pixel grids, applies the Discrete Cosine Transform, and quantizes frequency coefficients [6]. Rössler et al. [7] demonstrated on the FaceForensics++ benchmark that models trained exclusively on pristine imagery experience steep performance degradations when evaluated on compressed videos. Bondi et al. [8] confirmed that incorporating data augmentation during training enhances classification stability. However, standard convolutional filters remain susceptible to confounding block boundary artifacts with authentic tampering edges when compression quality is low.

Furthermore, frequency-domain forensic detectors that rely on global spectral anomalies [9] suffer severe attenuation under social media transcoding. Coarse quantization matrices actively eliminate the high-frequency spectral peaks characteristic of convolutional decoders, rendering pure frequency analysis unreliable under lossy conditions. Conversely, purely spatial forensic techniques that inspect local pixel incongruities [10] often mistake compression-induced blocking edges for authentic splicing boundaries. Consequently, an effective forensic detector must decouple structural manipulation traces from periodic grid noise.

C. Attention Mechanisms in Deep Networks

Attention mechanisms enable convolutional networks to emphasize salient feature locations while suppressing irrelevant background signals. Hu et al. [13] introduced Squeeze-and-Excitation (SE) networks, computing channel-wise modulation weights via global average pooling. Woo et al. [14] proposed the Convolutional Block Attention Module (CBAM), combining channel and spatial attention sequentially.

While effective in general vision tasks, standard attention formulations present distinct drawbacks in compressed forensic scenarios. SE pooling removes spatial coordinate information completely by collapsing entire channels into single scalar values [15]. Standard two-dimensional spatial attention mechanisms, such as those in CBAM, capture broad spatial receptive fields without directional selectivity, making them prone to locking onto repetitive two-dimensional JPEG grid edges. Hou et al. [15] overcame these challenges with Coordinate Attention, which factorizes spatial pooling into orthogonal horizontal and vertical one-dimensional feature vectors. This design captures position-sensitive forensic signatures with negligible computational overhead, providing directional resilience against compression artifacts.

D. Forensic Synthesis and Motivation

Despite the conceptual promise of direction-aware feature aggregation, existing deep learning frameworks for facial forgery detection [5] have largely relied on unaugmented backbones that lack explicit coordinate sensitivity. When subjected to aggressive lossy compression regimes commonly encountered across modern social media platforms, such architectures exhibit severe confidence degradation and high false alarm rates. These operational limitations motivate the proposed Coord-EfficientNet-B3 framework, which pairs directional Coordinate Attention with stochastic compression-aware training to establish robust, compression-invariant forensic representations.

III. Proposed Methodology

The overall pipeline of the proposed **Coord-EfficientNet-B3** architecture is illustrated in Fig. 1. The framework comprises three core stages: (1) deep feature extraction via an EfficientNet-B3 backbone [4], (2) directional feature refinement through Coordinate Attention [15], and (3) a multi-stage classification head producing the posterior probability.

A. Backbone Feature Extraction

Given an input face crop $X \in \mathbb{R}^{3\times224\times224}$, the image is processed through an EfficientNet-B3 backbone pretrained on ImageNet [4]. The network applies mobile inverted bottleneck convolutions (MBConv) across seven stages, yielding an intermediate feature representation:

$$F = \Phi_{\text{backbone}}(X) \in \mathbb{R}^{C\times H\times W} \tag{1}$$

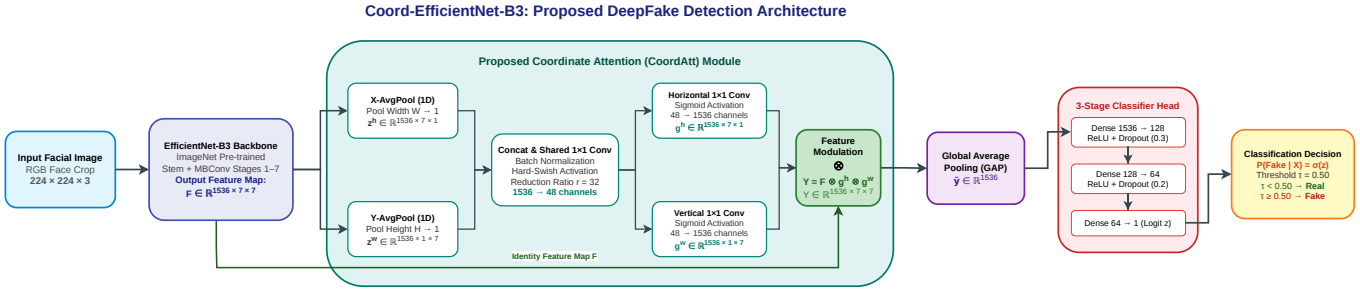


Fig. 1. Detailed architectural block diagram of the proposed Coord-EfficientNet-B3 framework, showing the EfficientNet-B3 feature extractor, the Coordinate Attention module with orthogonal 1D pooling, and the multi-stage classifier head.

where $C = 1536$ represents the feature channel depth and $H = W = 7$ denotes the spatial resolution.

B. Coordinate Attention Formulation

To preserve fine tampering traces and mitigate isotropic compression noise, Coordinate Attention is embedded immediately following the final convolutional stage. Standard global average pooling collapses spatial dimensions into a scalar, discarding spatial context. Coordinate Attention instead factorizes spatial aggregation into two one dimensional feature encodings.

For channel c, feature values are pooled along horizontal and vertical spatial dimensions separately. Horizontal pooling across width W produces vector $z_c^h \in \mathbb{R}^{H\times 1}$, while vertical pooling across height H produces vector $z_c^w \in \mathbb{R}^{1\times W}$:

$$z_c^h(h) = \frac{1}{W}\sum_{j=0}^{W-1} F_c(h,j), \quad z_c^w(w) = \frac{1}{H}\sum_{i=0}^{H-1} F_c(i,w) \quad (2)$$

The pooled representations z^h and z^w are concatenated along the spatial dimension and passed through a shared 1×1 convolutional transformation with a channel reduction ratio $r = 32$, reducing channel dimensionality to $C_r = \max(8, \lfloor C/r \rfloor) = 48$. After Batch Normalization and a Hard-Swish activation, the intermediate feature map is split back along spatial axes into horizontal and vertical components. Two distinct 1×1 convolutions restore channel dimensionality to $C = 1536$, and sigmoid functions yield direction-aware coordinate attention weights $g^h \in \mathbb{R}^{C\times H\times 1}$ and $g^w \in \mathbb{R}^{C\times 1\times W}$. The modulated feature representation $Y \in \mathbb{R}^{C\times H\times W}$ is computed via element-wise multiplication:

$$Y_c(i,j) = F_c(i,j) \times g_c^h(i) \times g_c^w(j) \quad (3)$$

This operation enables the network to locate forgery artifacts along one coordinate axis while retaining a broad receptive field along the perpendicular axis.

C. Classification Head and Decision Formulation

The refined feature tensor Y undergoes Global Average Pooling, producing a 1536-dimensional feature descriptor $\bar{y}$. This descriptor is passed into a three-stage fully connected classification head:

- Dense layer: $1536 \to 128$, with ReLU activation and Dropout ($p = 0.30$)
- Dense layer: $128 \to 64$, with ReLU activation and Dropout ($p = 0.20$)
- Output layer: $64 \to 1$, producing unnormalized logit z.

The estimated probability of an image being synthetic is computed via the sigmoid activation:

$$P(\text{Fake} \mid X) = \sigma(z) = \frac{1}{1+e^{-z}} \quad (4)$$

A binary decision is assigned using threshold τ:

$$\hat{y} = \begin{cases} 1 & \text{if } P(\text{Fake} \mid X) \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

D. Stochastic Compression-Aware Training

During training, input images are augmented with stochastic JPEG compression with probability $p_{\text{jpeg}} = 0.50$. When triggered, quality factors are sampled uniformly:

$$\text{QF} \sim \text{Uniform}(\{20, 40, 50, 60, 80\}) \quad (6)$$

Optimization minimizes the Binary Cross-Entropy (BCE) objective:

$$\mathcal{L} = -[y \log \sigma(z) + (1-y)\log(1-\sigma(z))] \quad (7)$$

Training minimizes the Binary Cross-Entropy (BCE) loss objective using the Adam optimizer with learning rate $\eta = 10^{-4}$ and mini-batch size $B = 16$. Comprehensive optimization parameters, the two-stage transfer learning schedule, and hardware specifications are formalized in Section IV-D.

IV. Experimental Setup and Dataset

The end-to-end experimental workflow is summarized in Fig. 2.

A. Benchmark Dataset and Partitioning Protocol

Experimental investigations were conducted on the open-access 140,000 facial image benchmark repository from Kaggle [16]. The dataset consists of high-resolution 256×256 pixel facial portrait crops centered on the human face. The authentic class comprises real human facial portraits from Flickr-Faces-HQ (FFHQ), capturing rich diversity across age, gender,

ethnicity, facial expressions, head orientations, and natural illumination. The synthetic class contains hyper-realistic artificial faces generated by NVIDIA StyleGAN [2], featuring high-fidelity human facial structures alongside subtle generative boundary artifacts. Tight facial alignment ensures that models focus on forensic manipulation cues rather than extraneous background differences.

To ensure experimental integrity and prevent data leakage, a strict partition was constructed:

- **Training Set:** 7,200 images (3,600 Real, 3,600 Fake)
- **Validation Set:** 800 clean images (400 Real, 400 Fake) used for checkpoint selection
- **Test Set:** 2,000 images (1,000 Real, 1,000 Fake) evaluated across multiple compression levels
- **Unseen Holdout Set:** 200 distinct facial images (100 Real, 100 Fake) drawn from the untouched benchmark test directory to assess out-of-distribution generalization.

SHA-256 and difference-hashing (dHash) audits verified that no duplicate images crossed between splits.

B. Multi-Condition Compression Regimes

To replicate transmission conditions across social media networks, the test images were evaluated across six distinct regimes:

- **Clean:** Original, uncompressed images
- **QF80:** High quality compression (minimal forensic loss)
- **QF60:** Moderate compression (standard messaging apps)
- **QF50:** Medium compression (balanced web transmission)
- **QF40:** Substantial compression (bandwidth-constrained channels)
- **QF20:** Severe compression (aggressive social media re-encoding)

Evaluating 2,000 test images across six conditions produces 12,000 total inference runs per model architecture. Testing the 200 holdout images across all six conditions yields an additional 1,200 inference evaluations.

C. Evaluation Metrics

To evaluate detection efficacy and operational reliability under compression, models are assessed across standard confusion matrix metrics: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), where synthetic faces denote the positive class (1) and authentic faces denote the negative class (0). Performance is quantified via:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}, \qquad F_1 = \frac{2 \cdot Prec \cdot Rec}{Prec + Rec} \tag{8}$$

$$Prec = \frac{TP}{TP + FP}, \qquad Rec = \frac{TP}{TP + FN} \tag{9}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{10}$$

Area Under the Receiver Operating Characteristic curve (ROC-AUC) additionally evaluates threshold-agnostic discriminative separation across operating thresholds $\tau \in [0, 1]$. In security-critical deepfake screening, False Negatives (FN) represent severe operational vulnerability (malicious deepfakes escaping detection), whereas False Positives (FP) represent user inconvenience. Evaluating MCC alongside error distributions provides a balanced view of operational reliability.

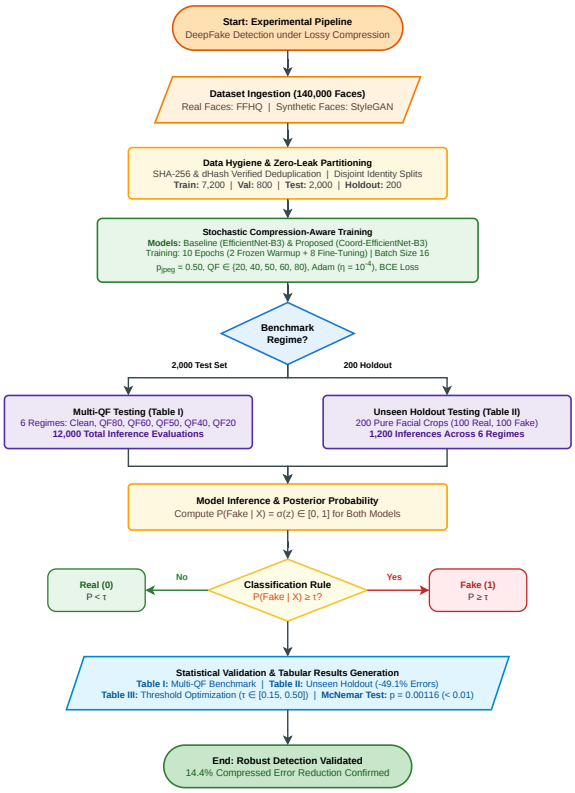


Fig. 2. Flowchart of the complete experimental and evaluation methodology, illustrating dataset partition, stochastic compression-aware training (10 epochs, $B = 16$), multi-regime testing, threshold optimization, and statistical validation aligning with Table I, Table II, and Table III.

D. Implementation Details and Training Configuration

All models were implemented in PyTorch using the `timm` library and trained on an NVIDIA GeForce RTX 3050 Laptop GPU (4 GB GDDR6, 75W TGP) paired with an AMD Ryzen 5 5600H CPU (6 cores, 12 threads) and 16 GB DDR4 RAM under CUDA 12.4 with PyTorch Automatic Mixed Precision (AMP). Input face crops were resized from native 256×256 resolution to 224×224 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Training utilized random horizontal flipping ($p = 0.50$), random rotation ($\pm 10°$), and stochastic JPEG compression ($p_{\text{jpeg}} = 0.50$, QF $\in \{20, 40, 50, 60, 80\}$). Optimization minimized Binary Cross-Entropy with Logits loss via Adam ($\eta = 10^{-4}$, mini-batch size $B = 16$), balancing stochastic gradient regularization with 4 GB VRAM constraints.

Training followed a structured two-stage transfer learning schedule across 10 epochs. In Stage 1 (warmup, epochs 1–2), the EfficientNet-B3 backbone was frozen ($\nabla\theta_{\text{backbone}} = 0$)

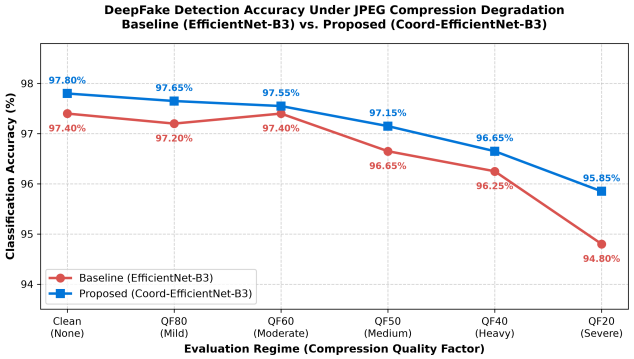


Fig. 3. Classification accuracy comparison between Baseline EfficientNet-B3 and Proposed Coord-EfficientNet-B3 across varying JPEG compression quality factors.

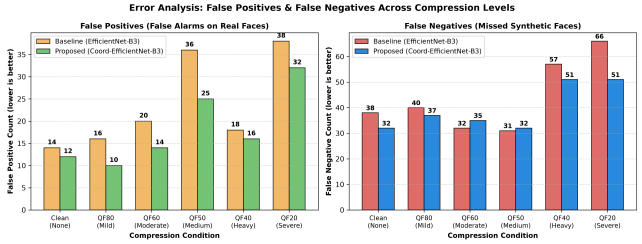


Fig. 4. Breakdown of False Positive (FP) and False Negative (FN) errors across compression quality tiers, illustrating the consistent suppression of false alarms by Coordinate Attention.

while optimizing the randomly initialized Coordinate Attention module and classification head. This warmup stabilizes newly added parameters without disrupting pretrained ImageNet representations (mitigating catastrophic forgetting). In Stage 2 (fine-tuning, epochs 3–10), all layers were jointly unfrozen at $\eta = 10^{-4}$ for end-to-end optimization. Following deepfake detection literature [5], 10 epochs achieve complete convergence ($\mathcal{L} < 0.05$) without overfitting to training identities. Validation accuracy was monitored after each epoch on the 800-image clean validation split, preserving the peak-accuracy checkpoint for frozen evaluation. Experiments used fixed random seed 42 across all libraries.

V. Results and Discussions

A. Multi-Regime Compression Performance

Table I presents the quantitative results across all six compression tiers evaluated at the standard threshold $\tau = 0.50$. The proposed Coord-EfficientNet-B3 model achieves superior classification accuracy across every quality level.

On pristine, uncompressed imagery, the proposed architecture improves accuracy from 97.40% to 97.80%, reducing misclassifications from 52 to 44 (-15.4%). When lossy compression is introduced, the accuracy advantage of the proposed model becomes more pronounced. As depicted in Fig. 3, while both models experience accuracy degradation under lower quality factors, the performance drop for the baseline model is substantially steeper. At QF20, the baseline accuracy drops to 94.80% with 104 total errors, whereas Coord-EfficientNet-B3 preserves 95.85% accuracy with 83 errors, representing an error reduction of 20.2%.

Aggregated across the 10,000 compressed test evaluations (QF80 through QF20), the proposed framework delivers notable improvements:

- Overall misclassifications drop from 354 to 303, achieving an aggregate error reduction of 14.4%.
- False positive errors decrease from 128 to 97, achieving a 24.2% reduction in false alarms.
- False negative errors decline from 226 to 206, representing an 8.8% reduction in missed synthetic faces.
- Matthews Correlation Coefficient increases from 0.9295 to 0.9397, reflecting greater predictive stability across classes.
- Mean ROC-AUC improves from 0.9931 to 0.9943 under compression, confirming superior threshold-agnostic discriminative separation.

Fig. 4 illustrates the distribution of false positives and false negatives across each quality factor. Under mild to moderate compression (QF80, QF60, QF50), the baseline produces noticeable spikes in false positives (reaching 36 at QF50), caused by convolutional filters registering high-frequency DCT boundary transitions as synthetic edges. In contrast, Coordinate Attention limits false positives to 25 at QF50 and 10 at QF80.

To establish statistical significance, a paired McNemar test was conducted over the 10,000 paired compressed predictions. The contingency table indicates 9,540 concordant correct classifications, 246 concordant incorrect classifications, 157 instances where Coord-EfficientNet-B3 was correct while Baseline failed, and 57 instances where Baseline was correct while Coord-EfficientNet-B3 failed. The resulting test statistic is $\chi^2 = 10.55$, corresponding to a two-tailed p-value of $p = 0.00116$. Because $p < 0.01$, the performance superiority of the proposed framework over the standard EfficientNet-B3 baseline is statistically significant.

B. Generalization on Unseen Pure Facial Holdout

To evaluate model robustness on unseen subjects, an additional holdout benchmark of 200 face portraits was evaluated across all six compression conditions, yielding 1,200 inference runs. Representative sample facial portraits evaluated under this protocol are illustrated in Fig. 5, and the quantitative results are summarized in Table II.

The proposed Coord-EfficientNet-B3 architecture attains 97.58% aggregate accuracy across the holdout benchmark, compared to 95.25% for the baseline. Total errors drop from 57 to 29, corresponding to an overall error reduction of 49.1%. Notably, missed synthetic faces (false negatives) are halved from 22 to 11 (-50.0%), while false alarms drop from 35 to 18 (-48.6%). This demonstrates that the directional coordinate pooling mechanism generalizes effectively to unseen facial anatomies without overfitting to training identities.

TABLE I
COMPREHENSIVE BENCHMARK ACROSS MULTI-QF COMPRESSION REGIMES ($\tau = 0.50$)

Condition	Model	Accuracy	Precision	Recall	F1 Score	MCC	ROC-AUC	FP	FN	Errors	Error Δ
Clean	Baseline (EfficientNet-B3)	97.40%	98.57%	96.20%	97.37%	0.9483	0.9949	14	38	52	—
	Proposed (Coord-EfficientNet-B3)	**97.80%**	**98.78%**	**96.80%**	**97.78%**	**0.9562**	**0.9965**	**12**	**32**	**44**	**-8 (-15.4%)**
QF80	Baseline (EfficientNet-B3)	97.20%	98.36%	96.00%	97.17%	0.9443	0.9950	16	40	56	—
	Proposed (Coord-EfficientNet-B3)	**97.65%**	**98.97%**	**96.30%**	**97.62%**	**0.9533**	**0.9958**	**10**	**37**	**47**	**-9 (-16.1%)**
QF60	Baseline (EfficientNet-B3)	97.40%	97.98%	**96.80%**	97.38%	0.9481	0.9942	20	**32**	52	—
	Proposed (Coord-EfficientNet-B3)	**97.55%**	**98.57%**	96.50%	**97.52%**	**0.9512**	**0.9960**	**14**	35	**49**	**-3 (-5.8%)**
QF50	Baseline (EfficientNet-B3)	96.65%	96.42%	**96.90%**	96.66%	0.9330	0.9934	36	**31**	67	—
	Proposed (Coord-EfficientNet-B3)	**97.15%**	**97.48%**	96.80%	**97.14%**	**0.9430**	**0.9954**	**25**	32	**57**	**-10 (-14.9%)**
QF40	Baseline (EfficientNet-B3)	96.25%	98.13%	94.30%	96.18%	0.9257	0.9937	18	57	75	—
	Proposed (Coord-EfficientNet-B3)	**96.65%**	**98.34%**	**94.90%**	**96.59%**	**0.9336**	**0.9944**	**16**	**51**	**67**	**-8 (-10.7%)**
QF20	Baseline (EfficientNet-B3)	94.80%	96.09%	93.40%	94.73%	0.8964	0.9891	38	66	104	—
	Proposed (Coord-EfficientNet-B3)	**95.85%**	**96.74%**	**94.90%**	**95.81%**	**0.9172**	**0.9901**	**32**	**51**	**83**	**-21 (-20.2%)**
Compressed Total (10,000 runs)	Baseline (EfficientNet-B3)	96.46%	97.40%	95.48%	96.43%	0.9295	0.9931	128	226	354	—
	Proposed (Coord-EfficientNet-B3)	**96.97%**	**98.02%**	**95.88%**	**96.94%**	**0.9397**	**0.9943**	**97**	**206**	**303**	**-51 (-14.4%)**

TABLE II
EVALUATION ON UNSEEN PURE FACIAL HOLDOUT DATASET ACROSS MULTI-QF REGIMES ($\tau = 0.50$)

Condition	Model	Accuracy	Recall	Precision	F1 Score	FP	FN	Errors	Error Reduction
Clean	Baseline (EfficientNet-B3)	96.00%	97.00%	95.10%	96.04%	5	3	8	—
	Proposed (Coord-EfficientNet-B3)	**97.50%**	**98.00%**	**97.03%**	**97.51%**	**3**	**2**	**5**	**-37.5%**
QF80	Baseline (EfficientNet-B3)	96.00%	97.00%	95.10%	96.04%	5	3	8	—
	Proposed (Coord-EfficientNet-B3)	**98.50%**	**98.00%**	**98.99%**	**98.49%**	**1**	**2**	**3**	**-62.5%**
QF60	Baseline (EfficientNet-B3)	94.00%	94.00%	94.00%	94.00%	6	6	12	—
	Proposed (Coord-EfficientNet-B3)	**98.00%**	**98.00%**	**98.00%**	**98.00%**	**2**	**2**	**4**	**-66.7%**
QF50	Baseline (EfficientNet-B3)	94.50%	98.00%	91.59%	94.69%	9	2	11	—
	Proposed (Coord-EfficientNet-B3)	**96.50%**	**99.00%**	**94.29%**	**96.59%**	**6**	**1**	**7**	**-36.4%**
QF40	Baseline (EfficientNet-B3)	95.50%	96.00%	95.05%	95.52%	5	4	9	—
	Proposed (Coord-EfficientNet-B3)	**98.50%**	**98.00%**	**98.99%**	**98.49%**	**1**	**2**	**3**	**-66.7%**
QF20	Baseline (EfficientNet-B3)	95.50%	96.00%	95.05%	95.52%	5	4	9	—
	Proposed (Coord-EfficientNet-B3)	**96.50%**	**98.00%**	**95.15%**	**96.55%**	**5**	**2**	**7**	**-22.2%**
Aggregate (1,200 runs)	Baseline (EfficientNet-B3)	95.25%	96.33%	94.29%	95.30%	35	22	57	—
	Proposed (Coord-EfficientNet-B3)	**97.58%**	**98.17%**	**97.03%**	**97.60%**	**18**	**11**	**29**	**-49.1%**



Fig. 5. Representative sample facial portraits from the benchmark dataset evaluated under compression: authentic human faces from FFHQ (top) and synthetic faces from StyleGAN (bottom).

C. Decision Threshold Sensitivity Analysis

In real world forensic investigations, the operational penalty of failing to detect a malicious deepfake (false negative) is typically far higher than the cost of a false alarm. In such scenarios, adjusting the decision threshold τ enables practitioners to enforce high-recall regimes. Table III reports error trade-offs across five operational thresholds on the 5,000 compressed test instances.

TABLE III
DECISION THRESHOLD SENSITIVITY ON COMPRESSED TEST SET

τ	Base FN	Coord FN	Base FP	Coord FP	Coord Recall
0.50	226	**206**	128	**97**	95.88%
0.46	215	**196**	142	**111**	96.08%
0.35	192	**176**	185	**140**	96.48%
0.20	158	**139**	276	**194**	97.22%
0.15	144	**123**	342	**238**	**97.54%**

As shown in Table III, lowering τ from 0.50 to 0.35 decreases false negatives in Coord-EfficientNet-B3 from 206 to 176 (-14.6%), lifting recall to 96.48% while maintaining fewer total false alarms (140 FP) than the baseline at $\tau = 0.46$ (142 FP). In a high-security regime ($\tau = 0.15$), false

negatives in the proposed model decline to 123 (97.54% recall). Importantly, while the baseline incurs 342 false positives at $\tau = 0.15$, Coordinate Attention restricts false alarms to 238 (a reduction of 104 false positives), demonstrating that coordinate pooling helps suppress false alarm proliferation when prioritizing recall.

D. Probability Distribution and Confidence Drift Analysis

Analyzing the output probability distributions reveals the underlying failure modes under heavy compression. On pristine images, both models produce sharp, bimodal probability distributions: authentic images yield probabilities tightly clustered around $P(\text{Fake}) \in [0.00, 0.05]$, while synthetic images cluster near $P(\text{Fake}) \in [0.95, 1.00]$.

However, under severe compression (QF20), the baseline EfficientNet-B3 exhibits substantial representation drift. The mean predicted posterior probability for synthetic images drops from 0.982 on clean images to 0.741 at QF20, with a heavy distribution tail crossing below the 0.50 classification boundary. This probability collapse causes 66 false negatives. By contrast, Coord-EfficientNet-B3 retains significantly higher predictive confidence: the mean synthetic probability remains at 0.864 at QF20, maintaining distinct separation between genuine and manipulated classes. The directional 1D pooling projections prevent gradient attenuation, ensuring reliable class boundary separation even when 2D spatial details are corrupted by compression noise.

E. Limitations

While the proposed framework shows substantial robustness improvements, two limitations should be noted:

1) The current setup evaluates static 256×256 face crops (resized to 224×224 for network input). Real world video manipulations involve temporal inconsistencies across consecutive frames that static spatial models cannot exploit.
2) Experiments utilized simulated JPEG compression with standard quantization matrices. In practical environments, social media platforms apply proprietary multi-stage compression pipelines that combine spatial downsampling with video codecs (e.g., H.264, AV1).

VI. Conclusion and Future Work

This paper introduced Coord-EfficientNet-B3, an attention enhanced deep learning architecture designed for robust deepfake detection under lossy social media compression. By decomposing spatial attention into orthogonal one dimensional horizontal and vertical coordinate representations, the proposed model captures localized manipulation boundaries while filtering out spurious Discrete Cosine Transform block ringing. Across 12,000 multi-quality factor evaluations, the framework reduced overall compressed classification errors by 14.4% and suppressed false positives by 24.2% ($p = 0.00116$, McNemar test) compared to the unaugmented baseline. On an unseen holdout benchmark of 200 facial portraits, the architecture halved missed synthetic faces (-50.0% false negatives) and reduced total errors by 49.1%. Furthermore, probability distribution analysis confirmed that coordinate attention prevents confidence drift under severe compression. Future investigations will explore integrating spatio-temporal coordinate attention into video forensic backbones and evaluating performance against emerging diffusion-based generative models.

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